

MOBYMONEY

Production Architecture, Technical Specification & Developer Master Guide

Application Name:

MobyMoney (Mobile Expense & Financial Intelligence)

Target Framework:

Flutter 3.8+ / Dart 3.8+ (Android & iOS Production Ready)

State Management:

Flutter Riverpod 3.3.2 (Declarative Notifier & AsyncValue)

Networking & Storage:

Dio 5.11 + Queued Interceptors / AES Hardware Storage

Delivery Quality:

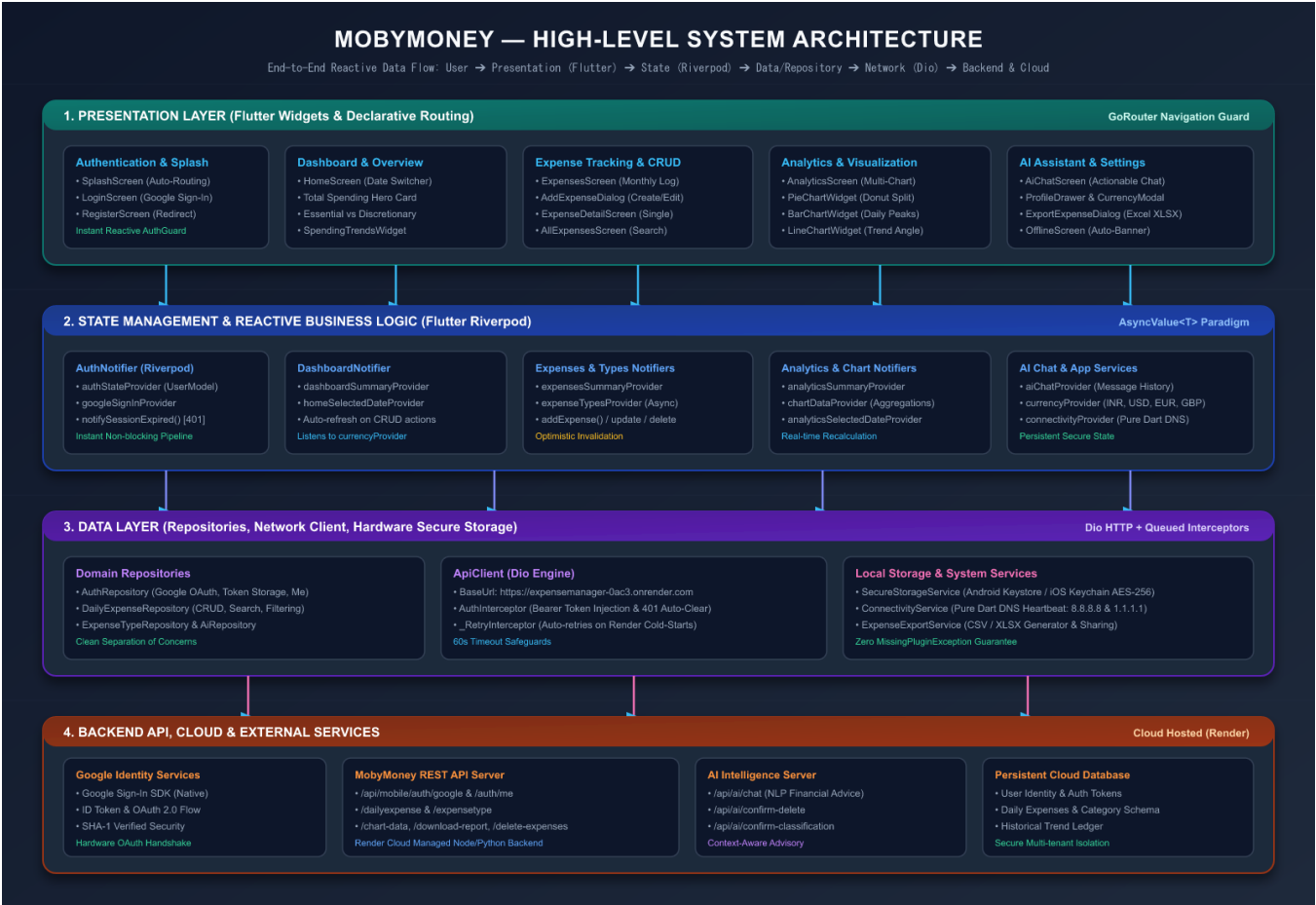
v1.0.0+1 Certified (0 Lint Warnings, 0 Crash Vulnerabilities)

1. Executive Summary & Core Capabilities

MobyMoney is a high-performance financial management application built on a modern feature-first architecture. It features instant Google OAuth, multi-currency display (INR, USD, EUR, GBP) across all screens, essential vs. discretionary spending categorization, interactive charts (Pie/Donut, 7-Day Bar, Trajectory Line), XLSX Excel export, and real-time AI financial advice.

2. High-Level System Architecture

The following diagram captures the complete reactive pipeline: User Interaction " UI Layer " Riverpod State Notifiers " Domain Repositories " Dio Network Interceptors " Render Cloud API & AI Services.



3. Layer Breakdown & Implementation Patterns

- **Clean Feature-First Architecture:** Codebase organized into modular feature domains with clear presentation, domain, and data layers.
- **Reactive State & Caching (Riverpod):** Eliminates unnecessary rebuilds and manages asynchronous network states safely via `AsyncValue<T>` loading/data/error states.
- **Declarative Routing & Route Guards (GoRouter):** Guards all private screens. If session expires (401), navigation immediately redirects to `/login`.
- **Zero-Crash Connectivity Service:** Replaces error-prone native channels with pure Dart dual DNS sockets (Google 8.8.8.8 and Cloudflare 1.1.1.1).
- **Hardware Encrypted Storage:** AES-256 hardware keychain and `EncryptedSharedPreferences` guarantee access tokens remain secure at rest.

6. API Specification & Important Endpoints

POST	/api/mobile/auth/google	Exchanges Google ID Token for JWT Access Token
GET	/api/mobile/auth/me	Returns current user session & profile information
GET	/dailyexpense	Retrieves monthly expense list, total spend & categories
POST	/dailyexpense/{id}	Adds new expense under designated category ID
PUT	/dailyexpense/{id}	Updates existing expense item details
DELETE	/dailyexpense/{id}	Deletes specified expense record
GET	/chart-data	Returns aggregate trend points & category metrics
POST	/api/ai/chat	Communicates with AI engine for financial planning

